E-Commerce Website - Complete Project Plan

Roman Urdu documentation | Stack: Next.js, Tailwind, shadcn/ui, Node.js, Express.js, Supabase | Version 1.0

1. Project Overview

Yeh ek full-stack e-commerce website hai jisme **sirf 1 Admin** hoga aur baqi sab users **Customer** honge.

- **Admin**: products ki CRUD, orders manage karna, CMS portal se website ka content control karna, coupons, customers, reports.
- **Customer**: products dekhna, cart/wishlist, order karna, order track karna, delivery ke baad review dena, profile aur addresses manage karna.
- **Theme**: Blue, White aur Black, modern gradients ke sath (details Section 9 mein).

Goals

- Fast, SEO-friendly aur mobile-first website.
- Admin ke liye aasaan dashboard jahan code likhe baghair content change ho sake.
- Secure system: sirf ek admin, baqi sab customers.
- Scalable architecture taake baad mein features add ho saken.

2. Roles aur Permissions

Feature	Guest	Customer	Admin
Products / categories dekhna	Yes	Yes	Yes
Search aur filter	Yes	Yes	Yes
Cart	Yes (local)	Yes (saved)	No
Checkout aur order	No	Yes	No
Order tracking	No	Apne orders	Sab orders

Feature	Guest	Customer	Admin
Review dena	No	Sirf delivered product par	Moderate kar sakta hai
Product CRUD	No	No	Yes
CMS control	No	No	Yes
Users / reports dekhna	No	No	Yes

Single Admin ka rule

- Public signup se hamesha `role = 'customer'` banega.
- Admin account **manually** Supabase se seed hoga (ek baar), website par admin signup ka koi route nahi hoga.
- Database mein unique index laga denge taake 2nd admin ban hi na sake.

3. Features List

3.1 Customer Features

1. Register / Login / Logout (Email + password, Google login optional)
2. Email verification aur Forgot / Reset password
3. Home page: hero banner, featured products, categories, offers
4. Product listing: search, filters (category, price, brand, rating), sorting, pagination
5. Product detail: image gallery, variants (size/color), stock status, reviews, related products
6. Cart: quantity update, remove, coupon apply
7. Wishlist
8. Checkout: address select/add, shipping method, payment method, order summary
9. Order placed email confirmation
10. My Orders: list, detail, status timeline, invoice download, cancel (jab tak Pending ho)
11. Order tracking: Pending > Confirmed > Processing > Shipped > Delivered
12. Reviews aur ratings (sirf Delivered order ke baad, 1 product par 1 review)
13. Profile: naam, phone, password change, saved addresses
14. Static pages: About, Contact, FAQ, Privacy Policy, Terms, Return Policy

15. Contact form aur newsletter subscribe
16. Dark mode toggle (optional)

3.2 Admin Features

1. Secure admin login
2. Dashboard: total sales, orders, customers, low stock alerts, charts
3. **Products CRUD**: add, edit, delete (soft delete), multiple images, variants, price, discount price, stock, SKU, SEO fields, publish/draft
4. **Categories / Brands CRUD**
5. **Orders management**: list, filter, status update, payment status, tracking number, invoice print, cancel/refund note
6. **Customers**: list, order history, block/unblock
7. **Reviews moderation**: approve, hide, delete
8. **Coupons / Discounts** management
9. **CMS Portal** (Section 4.3 mein detail)
10. **Settings**: store name, logo, currency, shipping charges, tax, payment methods, contact info
11. **Reports**: sales by date, top products, export CSV
12. Inventory: stock update aur low-stock warning
13. Notifications: naya order aane par email / in-app alert

3.3 CMS Portal (Admin)

Admin yeh sab bina developer ke change kar sake:

- **Hero banners / sliders**: image, heading, button text, link, order, active/inactive
- **Homepage sections**: featured products, new arrivals, categories ka order on/off
- **Static pages editor**: About, FAQ, Privacy, Terms (rich text editor)
- **Navigation menu aur footer links**
- **Announcement bar / promo strip**
- **Site settings**: logo, favicon, theme text, social links, contact details
- **SEO**: meta title, description, OG image per page aur per product
- **Blog** (optional phase 2)

4. Tech Stack aur Architecture

Layer	Technology	Kaam
Frontend	Next.js (App Router), JavaScript	UI, SSR/SSG, SEO, routing
Styling	Tailwind CSS + shadcn/ui	Components aur theme
Backend API	Node.js + Express.js	Business logic, REST API, validation
Database	Supabase (PostgreSQL)	Data storage
Auth	Supabase Auth (JWT)	Login/signup, sessions
File storage	Supabase Storage	Product images, banners, logo
Forms/validation	React Hook Form + Zod	Frontend + backend validation
Data fetching	TanStack Query (React Query)	Caching, loading states
State	Zustand	Cart, UI state
Rich text	Tiptap	CMS pages editor
Charts	Recharts	Admin dashboard
Email	Resend (ya Nodemailer)	Order emails, reset links
Payments	Cash on Delivery + Stripe (card)	Pakistan ke liye COD sab se zaroori hai
Security	Helmet, CORS, express-rate-limit	API protection
Hosting	Vercel (frontend), Render/Railway (Express), Supabase Cloud	Deployment

4.1 Architecture Diagram (Text)

```
[ Browser: Customer / Admin ]
          |
   Next.js App (Vercel)  -- SEO pages, UI, Supabase Auth login
          |  (JWT token bhejta hai)
   Express.js API (Render/Railway)
     - JWT verify middleware
     - Role check (admin/customer)
     - Business logic, validation
          |
   Supabase: PostgreSQL + Storage + Auth
```

4.2 Auth Flow

1. User Next.js se Supabase Auth se login karta hai, JWT milta hai.
2. Frontend har API request ke header mein `Authorization: Bearer <token>` bhejta hai.
3. Express middleware token verify karta hai aur `profiles` table se role check karta hai.
4. Admin routes par `requireAdmin` middleware lagega, warna 403 error.
5. Express **service role key** use karega jo sirf server par rahegi, kabhi frontend par nahi.

4.3 Rendering Strategy

- Home, product list, product detail: **SSR/ISR** (SEO ke liye)
- Cart, checkout, account, admin: **Client components** with protected routes
- Next.js `middleware.js` se `/admin/*` aur `/account/*` routes protect honge.

5. Database Design (Supabase PostgreSQL)

Tables ki list

`profiles`, `addresses`, `categories`, `brands`, `products`, `product_images`, `product_variants`, `cart_items`, `wishlist`, `orders`, `order_items`, `order_status_history`, `payments`, `reviews`, `coupons`, `banners`, `cms_pages`, `menu_items`, `site_settings`, `contact_messages`, `newsletter_subscribers`

Main tables ki SQL (sample)

```
-- Roles
create type user_role as enum ('customer', 'admin');
create type order_status as enum

('pending','confirmed','processing','shipped','delivered','cancelled','return
ed');
create type payment_status as enum ('unpaid','paid','refunded','failed');

-- Profiles (auth.users se linked)
create table profiles (
  id uuid primary key references auth.users(id) on delete cascade,
  full_name text,
  phone text,
  role user_role not null default 'customer',
  is_blocked boolean default false,
  created_at timestamptz default now()
);
-- Sirf ek admin allowed
create unique index only_one_admin on profiles (role) where role = 'admin';

create table categories (
  id uuid primary key default gen_random_uuid(),
  name text not null,
  slug text unique not null,
  image_url text,
  parent_id uuid references categories(id),
  is_active boolean default true
);

create table products (
  id uuid primary key default gen_random_uuid(),
  title text not null,
  slug text unique not null,
  description text,
  category_id uuid references categories(id),
  brand_id uuid references brands(id),
  price numeric(10,2) not null,
  discount_price numeric(10,2),
  sku text unique,
  stock int not null default 0,
  is_published boolean default false,
```

```
  is_featured boolean default false,
  avg_rating numeric(2,1) default 0,
  review_count int default 0,
  meta_title text,
  meta_description text,
  deleted_at timestamptz,
  created_at timestamptz default now()
);

create table orders (
  id uuid primary key default gen_random_uuid(),
  order_number text unique not null,
  user_id uuid references profiles(id),
  status order_status default 'pending',
  payment_method text not null,      -- 'cod' | 'card'
  payment_status payment_status default 'unpaid',
  subtotal numeric(10,2),
  shipping_fee numeric(10,2),
  discount numeric(10,2) default 0,
  total numeric(10,2),
  shipping_address jsonb not null,   -- snapshot
  tracking_number text,
  created_at timestamptz default now()
);

create table order_items (
  id uuid primary key default gen_random_uuid(),
  order_id uuid references orders(id) on delete cascade,
  product_id uuid references products(id),
  title text, unit_price numeric(10,2), quantity int
);

create table reviews (
  id uuid primary key default gen_random_uuid(),
  product_id uuid references products(id) on delete cascade,
  user_id uuid references profiles(id),
  order_id uuid references orders(id),
  rating int check (rating between 1 and 5),
  comment text,
  is_visible boolean default true,
  created_at timestamptz default now(),
  unique (product_id, user_id)       -- 1 user = 1 review per product
);
```

Important database rules

- Order place hone par **address aur price ka snapshot** save hoga, taake baad mein product price change ho to purani order na bigray.
- Stock kam karna aur order banana **ek transaction** (Postgres function / RPC) mein hoga taake overselling na ho.
- Review sirf tab allowed jab user ke paas us product ka `delivered` order ho (backend check karega).
- Review save hone par `products.avg_rating` aur `review_count` trigger se update honge.
- Product delete **soft delete** hoga (`deleted_at`), kyun ke purane orders us se linked hain.

6. Security Plan

1. **Row Level Security (RLS)** har table par ON ho. Customer sirf apna data dekh sake (cart, orders, addresses).
2. Admin ke liye policies `profiles.role = 'admin'` check karen.
3. Express mein har request par Zod validation aur sanitization.
4. `helmet` , strict `CORS` (sirf apni website ka domain), `express-rate-limit` (login aur contact form par khaas).
5. Passwords Supabase handle karta hai (bcrypt), khud store nahi karne.
6. Image upload par file type aur size limit (max 5MB, sirf jpg/png/webp).
7. `.env` secrets git mein commit nahi honge. Service role key sirf backend par.
8. HTTPS everywhere, secure cookies, XSS protection (CMS HTML ko sanitize karein, DOMPurify).
9. Admin actions ka **audit log** (kis ne kab kya change kiya).
10. Payment webhook signature verify karna (Stripe).

7. REST API Endpoints (Express.js)

Base URL: `/api/v1`

Public

Method	Route	Kaam
GET	`/products`	List (query: search, category, price, sort, page)
GET	`/products/:slug`	Product detail + reviews
GET	`/categories`	Categories list
GET	`/cms/home`	Banners, sections, settings
GET	`/cms/pages/:slug`	Static page content
POST	`/contact`	Contact form
POST	`/newsletter`	Subscribe

Customer (login zaroori)

Method	Route	Kaam
GET/PUT	`/me`	Profile dekhna / update
CRUD	`/me/addresses`	Addresses
GET/POST/DELETE	`/wishlist`	Wishlist
GET/POST/PUT/DELETE	`/cart`	Cart
POST	`/coupons/validate`	Coupon check
POST	`/orders`	Order place karna
GET	`/orders`	Mere orders
GET	`/orders/:id`	Order detail + tracking timeline
PATCH	`/orders/:id/cancel`	Cancel (sirf pending)
POST	`/products/:id/reviews`	Review add (delivered ke baad)
PUT/DELETE	`/reviews/:id`	Apna review edit/delete

Admin (`requireAdmin`)

Method	Route	Kaam
GET	`/admin/dashboard`	Stats aur charts
CRUD	`/admin/products`	Product management
POST	`/admin/products/:id/images`	Image upload
CRUD	`/admin/categories` , `/admin/brands`	Taxonomy
GET	`/admin/orders`	Sab orders (filters)
PATCH	`/admin/orders/:id/status`	Status + tracking number
GET/PATCH	`/admin/customers`	List, block/unblock
GET/PATCH/DELETE	`/admin/reviews`	Moderation
CRUD	`/admin/coupons`	Coupons
CRUD	`/admin/cms/banners`	Hero banners
CRUD	`/admin/cms/pages`	Static pages
PUT	`/admin/cms/menu`	Navigation/footer
PUT	`/admin/settings`	Site settings
GET	`/admin/reports/sales`	Reports + CSV export

Standard response format

```
{ "success": true, "data": { }, "message": "OK" }
{ "success": false, "error": "Validation failed", "details": [ ] }
```

8. Folder Structure

Frontend (Next.js)

```
frontend/
  app/
    (store)/              # public + customer pages
      page.js             # Home
      products/page.js
      products/[slug]/page.js
      cart/page.js
      checkout/page.js
      account/orders/page.js
      account/orders/[id]/page.js
      account/profile/page.js
      wishlist/page.js
      [slug]/page.js      # CMS static pages
    (auth)/login, register, forgot-password
    admin/
      layout.js           # sidebar + guard
      page.js             # dashboard
      products/, orders/, customers/, reviews/
      coupons/, settings/
      cms/banners, cms/pages, cms/menu
  components/
    ui/                   # shadcn/ui components
    store/                # ProductCard, CartDrawer, Navbar...
    admin/                # DataTable, StatCard, Charts...
  lib/                    # supabase client, api helper, utils
  hooks/  store/(zustand)  middleware.js  tailwind.config.js
```

Backend (Express)

```
backend/
  src/
    config/        # env, supabase client
    middleware/    # auth, requireAdmin, validate, errorHandler
    routes/        # public, customer, admin route files
    controllers/   # request handlers
    services/      # business logic (orderService, stockService)
    validators/    # Zod schemas
    utils/         # email, slugify, logger
    app.js  server.js
  .env
```

9. UI/UX Design System

Colors (Blue / White / Black)

Naam	Hex	Istemal
Primary Blue	`#2563EB`	Buttons, links
Deep Blue	`#1E3A8A`	Hover, headings
Sky Blue	`#38BDF8`	Accents, gradient end
Black / Ink	`#0B0F1A`	Navbar, footer, dark sections
Soft White	`#F8FAFC`	Page background
Pure White	`#FFFFFF`	Cards
Gray text	`#64748B`	Secondary text
Success / Warning / Danger	`#16A34A` / `#F59E0B` / `#DC2626`	Status badges

Gradients

```
/* Hero / CTA buttons */
background: linear-gradient(135deg, #1E3A8A 0%, #2563EB 50%, #38BDF8 100%);

/* Dark premium section / footer */
background: linear-gradient(180deg, #0B0F1A 0%, #111B3A 100%);

/* Sale badge / highlights */
background: linear-gradient(90deg, #2563EB, #0B0F1A);

/* Subtle section background */
background: linear-gradient(180deg, #F8FAFC 0%, #E0ECFF 100%);
```

Design guidelines

- **Font**: Inter ya Poppins (headings ke liye Poppins).
- **Cards**: rounded-2xl, soft shadow, hover par halka lift + blue glow.
- **Navbar**: black/glass effect (backdrop-blur), search bar, cart icon with badge.
- **Hero**: blue gradient background, bara heading, white CTA button.
- **Product card**: image, discount badge, rating stars, price, quick add-to-cart.
- **Animations**: Framer Motion se smooth fade/slide (zyada nahi, performance ka khayal).
- **Responsive**: mobile-first (sm, md, lg, xl), mobile par bottom cart bar.
- **Admin panel**: dark sidebar (black), white content area, blue active states.
- **shadcn/ui components**: Button, Card, Dialog, Sheet (cart drawer), Table, Tabs, Select, Dropdown, Badge, Toast (Sonner), Skeleton, Form, Input, Checkbox, Calendar, Pagination.
- **Accessibility**: proper contrast, alt text, keyboard navigation, focus rings.

10. Order Flow

```
Customer checkout > Order [Pending] > Admin confirm [Confirmed]
 > [Processing] > [Shipped + tracking no.] > [Delivered]
                                          > Review unlock
Cancel: sirf Pending par customer kar sakta hai
Return: Delivered ke baad admin [Returned] mark kare
```

- Har status change `order_status_history` mein save hoga (timeline ke liye).

- Har status par customer ko **email** jayegi.
- COD order: Delivered par `payment_status = paid` hoga.
- Card order: Stripe webhook se `paid` hoga.

11. Development Phases aur Timeline

Phase	Kaam	Muddat (approx)
1. Setup	Repos, Next.js + Tailwind + shadcn, Express, Supabase project, env, ESLint/Prettier	2-3 din
2. Database + Auth	Tables, RLS, triggers, admin seed, login/register, middleware	4-5 din
3. Admin: Catalog	Dashboard layout, Categories, Products CRUD, image upload	5-6 din
4. Store Frontend	Home, listing, filters, detail page, search	6-7 din
5. Cart + Checkout	Cart, wishlist, addresses, coupons, order placement (transaction)	5-6 din
6. Orders	Customer tracking, admin order management, emails, invoice	4-5 din
7. Reviews	Review rules, moderation, rating aggregation	2-3 din
8. CMS Portal	Banners, pages editor, menu, settings, SEO	5-6 din
9. Reports + Polish	Charts, CSV export, animations, responsive fixes	3-4 din
10. Testing + Deploy	Testing, security review, deploy, domain, monitoring	4-5 din

Total approx: 6 se 8 hafte (1 developer, full time). Part-time mein 10-14 hafte.

12. Testing Plan

- **Unit tests**: Jest (services jaise price calculation, stock logic)
- **API tests**: Supertest + Jest (auth, orders, admin guard)
- **E2E tests**: Playwright (register > add to cart > checkout > track order > review)

- **Manual checklist**: mobile devices, slow internet, alag browsers
- **Security checks**: customer admin routes ko access na kar sake, RLS verify, rate limiting

13. Deployment Plan

Part	Platform
Next.js frontend	Vercel
Express API	Render / Railway / VPS
Database, Auth, Storage	Supabase Cloud
Domain + SSL	Namecheap / Cloudflare
Emails	Resend
Monitoring	Sentry (errors), Vercel Analytics, UptimeRobot
CI/CD	GitHub Actions (lint + test + auto deploy)

Environment Variables

```
# Frontend
NEXT_PUBLIC_SUPABASE_URL=
NEXT_PUBLIC_SUPABASE_ANON_KEY=
NEXT_PUBLIC_API_URL=

# Backend
SUPABASE_URL=
SUPABASE_SERVICE_ROLE_KEY=   # kabhi frontend par nahi
FRONTEND_URL=
RESEND_API_KEY=
STRIPE_SECRET_KEY=
STRIPE_WEBHOOK_SECRET=
```

14. Performance aur SEO

- `next/image` se images optimize, WebP format, lazy loading.
- ISR se product pages cache, `revalidate` jab admin product update kare.

- Dynamic `sitemap.xml`, `robots.txt`, meta tags, Open Graph, product JSON-LD schema.
- Database indexes: `products(slug)`, `products(category_id)`, `orders(user_id)`, full-text search index.
- Pagination har list par, kabhi sab data ek sath load nahi.

15. Future Enhancements (Phase 2)

- JazzCash / Easypaisa payment integration
- Courier API integration (TCS, Leopards, PostEx) auto tracking
- Blog module, product Q&A, recently viewed
- WhatsApp order notifications
- Multi-language (English + Urdu) aur RTL support
- PWA / mobile app (React Native)
- Abandoned cart emails, product recommendations

16. Aglay Qadam (Next Steps)

1. Is plan ko review karen aur jo cheez add/remove karni ho batayen.
2. Business details final karen: store ka naam, logo, currency (PKR), shipping charges, payment methods.
3. Supabase project banayen aur GitHub repos setup karen.
4. **Phase 1 se coding shuru karen**. Main aap ko step-by-step code (setup, database SQL, auth, admin panel) bana kar de sakta hoon.